

Chapter 1: Foundations of Free Software

INTRODUCTION:

Modern computing relies on fundamental concepts such as software, source code, and licenses, which determine how technologies are used, shared, and developed in the digital world.

Understanding the following concepts is essential to mastering the legal, technical and economic issues of free and open source software.

a) What is **software**?

Software is a set of programs and instructions that allow a computer or digital device to perform specific tasks. It serves as an interface between the user and the computer hardware.

The executable program is what the computer understands.

We distinguish between system software, such as Windows or Linux, and application software, such as Word, Excel, or a web browser.

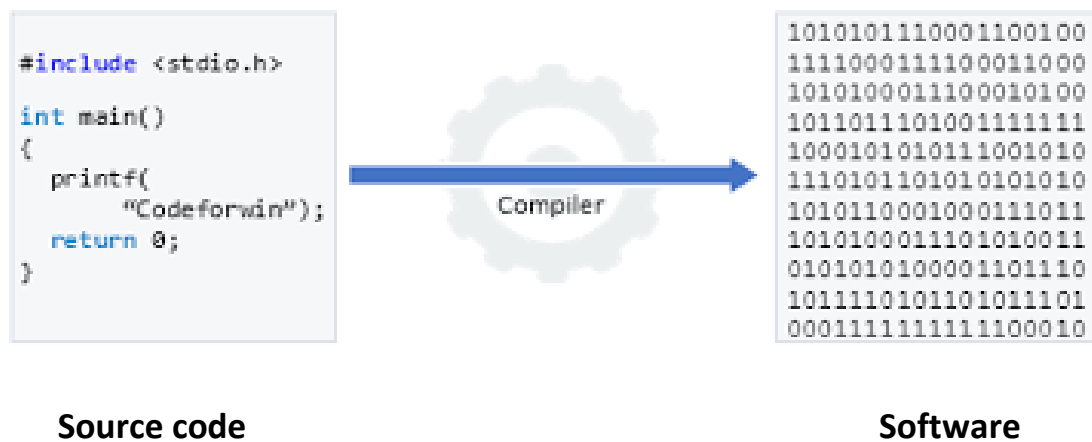
Without software, computer hardware is useless.

In short, software is the central element of any computer system.

b) What is **source code**?

Source code is the set of instructions written by developers in a understandable programming language. It forms the basis of the software and allows it to be understood, modified, and improved.

Source code is what humans can read and modify.



c) What is an **operating system**?

An operating system is the main software on a computer that manages hardware resources (processor, memory, peripherals) and allows other software to run. It forms the basis of how a computer system functions.

Note:

Linux is a free operating system.

Windows is a proprietary operating system.

d) What is **free software**?

Free software is software that guarantees its users four fundamental freedoms: to use the program, to study how it works, to modify it, and to redistribute it. These freedoms are ensured by free licenses.

e) What is **open-source software**?

The term open source refers to software whose source code is publicly accessible and can be studied, modified, and redistributed. It emphasizes collaboration, code quality, and technical innovation.

f) What is a **software license**?

A software license is a legal document that defines the rights and obligations related to the use, modification, and distribution of software. It specifies what the user is and is not allowed to do with the software.

1.1) History of the free and open source software movement:

◆ The Beginnings of Computing (1950s–1970s)

In the early days of computing, computers were very expensive and very rare. They were mainly found in universities and research centers.



At that time, computers were distributed with software specifically designed for them (whose value was secondary).

The source code was always provided and shared among researchers.

Cooperation was natural: everyone improved the work of others.

The 1960s and 70s were marked by hackers, most often brilliant students from the best American universities, who threw themselves passionately into the first stirrings of computing.

They spent nights working on their programs, waiting to be able to access a few hours of computer time, which was a scarce commodity. They shared their tips and programs within various clubs.

- Sharing was the norm.
- The concept of proprietary software was virtually nonexistent.

Note: The term "hacker" at that time did not have the same negative connotation as it does today: a hacker was then a programmer who was both passionate and gifted; they were the ones who laid the foundations

of modern computing, and many went on to create today's leading companies.

The values of hacker culture:

- | | |
|---------------------|------------------------------|
| -Curiosity | -Motivation |
| -Sense of challenge | -Sharing |
| -Sociability | - Mutual support, solidarity |
| -Freedom | -Knowledge and competence |
| -Respect | |

◆ Emergence of Proprietary Software (1970s–1980s)

With the arrival of computer companies, software became a source of profit. “The 1976 Homebrew Computer Club meeting could be considered a significant date in the split between frees and proprietary software. At this meeting, Bill Gates and Paul Allen presented a BASIC interpreter program they had written for the Altair 100, one of the first microprocessor computers.





Club members took the punched tape containing the program, duplicated it, and distributed it. Bill Gates, furious, wrote a now-famous letter entitled "Open Letter to Hobbyists," in which he explained that developers' work should be fairly compensated, and that if it isn't, innovation will be stifled. The reasoning is sound, and yet the future will show that it is also possible to develop large programs in open source.”[5]

Key points of the open letter written by Bill Gates in January 1976:

- ***Accusation of theft:*** Bill Gates denounces the sharing and reproduction of software without paying the developers, calling it "theft."
- ***Impact on innovation:*** According to him, free software discourages the development of professional software, because no one can invest time without compensation.
- ***Context:*** The letter was aimed at the hobbyist community who used Altair BASIC without a license.
- ***Reactions:*** It divided the community, with some criticizing Gates's vision, while others launched projects like Tiny BASIC to create free software.

This letter is considered a pivotal moment in the opposition between proprietary software and the free software movement.

Companies close the source code, prohibiting copying and modification.

Microsoft takes off through an agreement with IBM:
IBM sells PCs (personal computers) with Microsoft's operating system (MS-DOS), over which Microsoft retains the rights.

➤ **The software industry is born.**

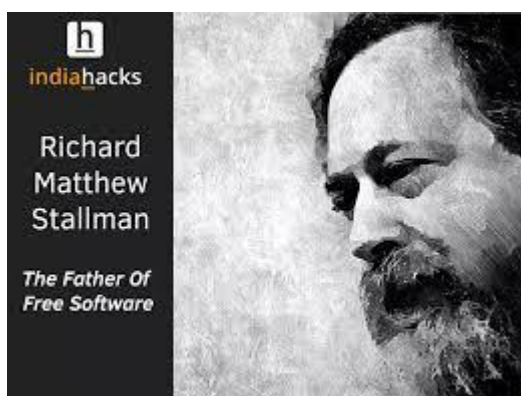
Proprietary software creates dependency: the user no longer controls their computer.

- Software becomes a means of control.
- Programmers lose their freedoms.

◆ **Birth of the Free Software Movement (1983)**

Richard Stallman, a programmer at MIT, rejected this situation. He considered depriving a user of freedom a social injustice. In 1983, he launched the GNU project to create a completely free operating system.

- Free software is an ethical response.



◆ **Founding of the Free Software Foundation (1985)**



In 1985, Stallman founded the Free Software Foundation.
It legally protects free software and created the GPL license.



The GPL guarantees that all free software remains free, even after modification.

➤ Freedom is protected by law.

◆ **The emergence of Linux (1991)**

Linus Torvalds, a Finnish student, created a free kernel: Linux.



By combining it with GNU tools, a complete system was created: GNU/Linux.



This was the first fully functional free operating system.

- Global collaboration became possible.

◆ Birth of the term “Open Source” (1998)

Some companies found Stallman's discourse too ideological.

The term Open Source was created to emphasize technical quality rather than ethics.



- Same practice, different philosophy.
- Creation of the Open Source Initiative (OSI), with an emphasis on:
 - quality
 - security
 - efficiency

- collaborative development

◆ **Global distribution (2000s)**

Today, free and open-source software is everywhere:
servers, the internet, smartphones, the cloud, artificial intelligence.

◆ **Current situation**

Free and open-source software is:

- everywhere
- essential
- strategic

◆ **Has free software won?**

One might think so if we look at the tools used by developers on their computers (editors, version control systems, compilers, libraries).

But if we look at it in terms of end-user freedom, we are far from it!

Most web and mobile applications depend on free software but do not give users any freedom.

GitHub is not free! (Even though free alternatives, such as GitLab and Gitea, exist.)

And Microsoft Word continues to dominate the market against LibreOffice.[5]

◆ **General Conclusion:**

- The free software movement was born from a fight for freedom.
- Open source has enabled mass adoption.
- Together, they have transformed global computing.